Canvas Quiz 3, Modules 4–5 (graded)

Module 4 - Drug Elimination and Clearance · Canvas quiz · 23 Sep 2026

Name _____ Date _____

Question 1 Which term describes the process by which drug is chemically converted in the body?

A. biotransformation
B. distribution
C. excretion
D. absorption

Question 2 If there is an increase in the total body clearance of an agent due to renal dysfunction, what is the expected change to the elimination half-life of that agent?

A. The elimination half-life will increase.
B. The elimination half-life will decrease.
C. The elimination half-life will not change.

Question 3 Which statement is true regarding the time needed to reach the maximum concentration following administration of an oral dose?

A. the time needed to reach the maximum concentration depends on the rate constants for absorption and elimination
B. the time needed to reach the maximum concentration increases if the dose is decreased
C. the time needed to reach the maximum concentration increases if the dose is increased
D. the time needed to reach the maximum concentration is independent of the rate constants for absorption and elimination

Question 4 Which of the following is true regarding the C_{max} following oral administration?

A. the rate of drug absorption equals the rate of drug elimination
B. the rate of drug elimination is faster than the rate of drug absorption
C. drug at the absorption site has become depleted
D. the rate of drug absorption is faster than the rate of drug elimination

Question 5 When does the maximum concentration of drug in the plasma occur following a single 1000-mg oral dose of an agent that has an oral bioavailability of 83%, apparent volume of distribution of 18 L, absorption half-life of 0.8 hours, and elimination half-life of 5.6 hours? Round to the nearest tenth (one decimal place).

Question 6 A 500-mg IV bolus dose of an antibiotic with an apparent volume of distribution of 23 L and half-life of 6.0 hr was administered to a 75 kg male volunteer. What is the expected AUC of this dose? Round to the nearest tenth (one decimal place).

Question 7 1,000 mg of a drug was given by IV bolus injection. The V_D is 19 L and the elimination half-life is 3 hours. Urine samples were collected for 48 hours and 713 mg of unchanged drug was recovered. What is the renal clearance of this drug? Round to the nearest hundredth (two decimal places).

Question 8 The kinetics following the oral administration of a single 500-mg dose of a medicinal agent best fits a one-compartment model described by the equation $C_p = 30(e^{-0.133t} - e^{-0.934t})$. What is the volume of distribution following the 500-mg dose if the oral bioavailability is 82%? Assume units of mcg/mL for C_p and hr for time. Round to the nearest tenth (one decimal place).

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Question 1 Worked by the drill

Answer: A. biotransformation

Biotransformation, also called metabolism, is chemical change of the drug into a metabolite. It is one of the two elimination processes; the other, excretion, removes the drug unchanged.

Question 2 Worked by the drill

Answer: B. The elimination half-life will decrease.

$t½ = \frac{0.693 \times V_D}{Cl}$. With V_D unchanged, a larger clearance means a shorter half-life. The item asks for the direction that follows from the stated change in clearance, whatever caused it.

The stem pairs renal dysfunction with an increase in clearance. Her keyed answer follows the stated increase in clearance; it is the Cl to t½ relation that is tested, not the usual direction of renal disease.

Question 3 Worked by the drill

Answer: A. the time needed to reach the maximum concentration depends on the rate constants for absorption and elimination

$t_{max} = \frac{\ln(k_a/k)}{k_a - k}$. Only the two rate constants appear; dose, F and V_D scale the curve without moving its peak in time.

Question 4 Worked by the drill

Answer: A. the rate of drug absorption equals the rate of drug elimination

At the peak the concentration is momentarily not changing, so drug is entering the body exactly as fast as it is leaving. That balance is what defines the maximum.

Question 5 Worked by the drill

Answer: 2.6 hr

WHAT THE STEM GIVES

D_0	1000-mg oral dose	*not needed: t_{max} does not depend on the dose*
F	83%	*not needed: F scales the height of the curve, not its timing*
V_D	18 L	*not needed: t_{max} has no volume in it*
absorption t½	0.8 hours	gives $k_a = \frac{0.693}{0.8} = 0.8663\ hr^{-1}$

t½	5.6 hours	gives $k = \frac{0.693}{5.6} = 0.1238\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max} = \frac{\ln(k_a / k)}{k_a - k}$$

The time to peak depends on the two rate constants only. The dose, the 83% bioavailability and the 18 L volume multiply the whole curve and cancel, so they are left alone.

ALGEBRA

$$k_a = \frac{0.693}{0.8\ hr} = 0.8663\ hr^{-1};\ k = \frac{0.693}{5.6\ hr} = 0.1238\ hr^{-1}$$

The t_{max} equation takes rate constants, not half-lives, so each half-life is converted with $k = \frac{0.693}{t½}$: k_a (the absorption rate constant) from the 0.8-hour absorption half-life and k (the elimination rate constant) from the 5.6-hour elimination half-life. 0.693 over hours gives hr^{-1} for both, so they can be subtracted.

ALGEBRA

$$t_{max} = \frac{\ln(0.8663 / 0.1238)}{0.8663 - 0.1238} = \frac{\ln 7.0}{0.7425\ hr^{-1}} = \frac{1.9459}{0.7425\ hr^{-1}} = 2.621\ hr$$

The ratio of the rate constants is the ratio of the half-lives the other way round, 5.6/0.8 = 7, so the logarithm is ln 7. Dividing by reciprocal hours leaves hours.

ROUND

2.6 hr

t_{max} = 2.621 hr rounds to 2.6 hr at one decimal place, as the stem asks. Her key is 2.6 with a margin of 3%, so 2.52 to 2.68 scores. The peak comes early relative to the 5.6-hour elimination half-life because absorption, with a 0.8-hour half-life, is the faster of the two processes.

> **DOES THE NUMBER MAKE SENSE?**
> Absorption half-life 0.8 hr against elimination 5.6 hr, 7 times slower: the peak comes a few absorption half-lives in and well before one elimination half-life of 5.6 hr: 2.6 hr.

Question 6

Worked by the drill

Answer: 188.2 (mg/L)hr

WHAT THE STEM GIVES

D_0	500-mg IV bolus	the numerator
V_D	23 L	in $Cl = kV_D = 0.1155 \times 23 = 2.657$ L/hr
t½	6.0 hr	gives $k = \frac{0.693}{6.0} = 0.1155\ hr^{-1}$
weight	75 kg	*not needed: dose and volume are already whole-patient values*

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{AUC} = \frac{D_0}{Cl} = \frac{D_0}{k \times V_D}$$

For an IV dose the whole dose is eliminated, and clearance is dose over AUC, so AUC is dose over clearance. Clearance is not given and is built from k and V_D. The 75 kg is not used because V_D is given in litres already.

ALGEBRA

$$k = \frac{0.693}{6.0\ \text{hr}} = 0.1155\ \text{hr}^{-1};\ Cl = 0.1155 \times 23\ \text{L} = 2.657\ \text{L/hr}$$

Cl (clearance) is not given, so it is built as $k \times V_D$: k (the elimination rate constant) = $\frac{0.693}{6.0\ \text{hr}}$ = $0.1155\ \text{hr}^{-1}$ from the half-life, and V_D (the volume of distribution) is 23 L. $\text{hr}^{-1} \times \text{L}$ gives L/hr: $0.1155 \times 23 = 2.657$ L/hr.

ALGEBRA

$$\text{AUC} = \frac{500\ \text{mg}}{2.657\ \text{L/hr}} = 188.2\ \text{(mg/L)hr}$$

mg divided by L/hr is mg·hr/L, which is a concentration times a time, the unit of an area under a concentration curve.

ROUND

188.2 (mg/L)hr

Her key is 188.2 with a margin of 3%, so 182.6 to 193.8 scores. The same answer comes from C_0/k: $C_0 = 500/23 = 21.74$ mg/L, and $21.74/0.1155 = 188.2$.

DOES THE NUMBER MAKE SENSE?

$C_0 = \frac{500}{23}$ is about 22 mg/L and AUC = $\frac{C_0}{k} = \frac{22}{0.1155}$, which also comes to about 188 (mg/L)hr, agreeing with the clearance route.

Question 7

Worked by the drill

Answer: 3.13 L/hr

WHAT THE STEM GIVES

D_0	1,000 mg	the numerator of AUC = $\frac{D_0}{kV_D}$
V_D	19 L	in $Cl = kV_D = 0.231 \times 19 = 4.389$ L/hr
t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ \text{hr}^{-1}$
collection	48 hours	*not needed for the number: 16 half-lives, so the urine holds all unchanged drug*
$D_u\infty$	713 mg	the numerator of $Cl_R = \frac{D_u\infty}{AUC}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_R = \frac{D_u\infty}{AUC}, \text{ with } AUC = \frac{D_0}{k \times V_D}$$

Renal clearance (Cl_R) is the amount excreted unchanged over the whole time course, $D_u\infty$, divided by the total AUC (area under the curve). The 48-hour collection is $\frac{48 \text{ hr}}{3 \text{ hr}} = 16$ half-lives, so essentially all the drug has been eliminated and 713 mg is the total excreted unchanged.

ALGEBRA

$$k = \frac{0.693}{3 \text{ hr}} = 0.231 \text{ hr}^{-1}; \quad AUC = \frac{1000 \text{ mg}}{0.231 \text{ hr}^{-1} \times 19 \text{ L}} = \frac{1000}{4.389} = 227.8 \text{ (mg/L)hr}$$

The total AUC is $\frac{D_0}{Cl}$, with $Cl = k \times V_D$. k (the elimination rate constant) = $\frac{0.693}{3 \text{ hr}} = 0.231$ hr^{-1} from the half-life, and V_D (the volume of distribution) is 19 L, so Cl = 0.231 × 19 = 4.389 L/hr. mg over L/hr leaves (mg/L)hr: $\frac{1000}{4.389} = 227.8$.

ALGEBRA

$$Cl_R = \frac{713 \text{ mg}}{227.8 \text{ (mg/L)hr}} = 3.13 \text{ L/hr}$$

Cl_R (renal clearance) = $\frac{D_u\infty}{AUC}$, where $D_u\infty$ is the 713 mg recovered unchanged: mg divided by (mg/L)hr leaves L/hr, the volume of plasma cleared by the kidney each hour. The same answer comes from $f_e \times Cl_T$, with $f_e = \frac{713}{1000} = 0.713$ the fraction excreted unchanged: 0.713 × 4.389 = 3.13 L/hr.

ROUND

3.13 L/hr

Cl_R = 3.13 L/hr at two decimal places, as the stem asks. Her key is 3.13 with a margin of 3%, so 3.04 to 3.22 scores. Renal clearance is 3.13 of the 4.389 L/hr total, which matches f_e = 0.713: the kidney accounts for about 71 per cent of elimination and the remainder is non-renal.

DOES THE NUMBER MAKE SENSE?

The kidney removed $\frac{713}{1000}$ of the dose, so renal clearance is that fraction of the total 4.389 L/hr, a little over 3 L/hr: 3.13 L/hr, below the total.

Question 8

Worked by the drill

Answer: 15.9 L

WHAT THE STEM GIVES

D_0	500-mg dose	in the numerator F D_0 k_a
coefficient	30	in the denominator 30 × (k_a − k)
k	0.133	the smaller exponent, elimination; in k_a − k

k_a	0.934	the larger exponent, absorption; in the numerator and in $k_a - k$
F	82%	written 0.82, in the numerator
units	mcg/mL for Cp and hr for time	mcg/mL = mg/L, so mg over mg/L gives litres

THE WORKING, ONE LINE AT A TIME

SETUP

The coefficient $30 = \dfrac{F \times D_0 \times k_a}{V_D \times (k_a - k)}$, so $V_D = \dfrac{F \times D_0 \times k_a}{30 \times (k_a - k)}$

The single oral dose equation is $C_p = [F\ D_0\ k_a / (V_D\ (k_a - k))]\ (e^{-kt} - e^{-k_a\ t})$. Matching it to the given equation identifies the coefficient and the two rate constants; the larger constant, 0.934, is k_a and the smaller, 0.133, is k, because in ordinary absorption k_a exceeds k.

UNIT

mcg/mL = mg/L, so a dose in mg and a coefficient in mcg/mL give V_D in L

The concentration unit decides the volume unit. With C_p in mg/L and the dose in mg, the volume comes out in litres without any further conversion.

ALGEBRA

$$V_D = \frac{0.82 \times 500\ \text{mg} \times 0.934\ \text{hr}^{-1}}{30\ \text{mg/L} \times (0.934 - 0.133)\ \text{hr}^{-1}} = \frac{382.9}{24.03} = 15.94\ \text{L}$$

The numerator 0.82×500 mg $\times 0.934$ $\text{hr}^{-1} = 382.9$ mg hr^{-1}, the denominator 30 mg/L $\times$ $(0.934 - 0.133)$ $\text{hr}^{-1} = 30 \times 0.801 = 24.03$ (mg/L) hr^{-1}. The hr^{-1} cancels top and bottom, and mg over mg/L leaves litres: $\dfrac{382.9}{24.03} = 15.94$ L.

ROUND

15.9 L

$V_D = 15.94$ L rounds to 15.9 L at one decimal place. Her key is 15.9 with a margin of 3%, so 15.4 to 16.4 scores. Leaving F out of the numerator gives $\dfrac{15.94}{0.82} = 19.4$ L, too large; swapping k_a and k makes $(k_a - k)$ negative and gives a negative volume.

DOES THE NUMBER MAKE SENSE?

The prefactor $\dfrac{F\ D_0\ k_a}{V_D(k_a - k)}$ is 30 mg/L and k_a over $(k_a - k)$ is a little above 1, so V_D is a little above $\dfrac{0.82 \times 500}{30}$, roughly 14 L: 15.9 L, above zero.